

Jungho Lee

RESEARCH SCIENTIST AT NAVER AI LAB

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RESEARCH INTERESTS

3D Computer Vision

- Geometric Foundation Models
- 3D Scene Representation
- 3D Neural Rendering

Physical AI

- World Foundation Models

EDUCATION

Yonsei University | College of Engineering

INTERATED M.S./PH.D IN ELECTRICAL AND ELECTRONIC ENGINEERING

- Image and Video Pattern Recognition Lab.
- Advisor: Prof. Sangyoun Lee

Seoul, South Korea

Sep. 2021 - Aug. 2026

Yonsei University | College of Engineering

B.S. IN ELECTRICAL & ELECTRONIC ENGINEERING

- 2-Year Military Service (2017-2019)

Seoul, South Korea

Mar. 2015 - Aug. 2021

Whimoon High School

Seoul, South Korea

Mar. 2010 - Feb. 2013

EXPERIENCE

NAVER AI Lab

RESEARCH SCIENTIST

- Generation Research

Seongnam, South Korea

Aug. 2026 - Present

NAVER Cloud

RESEARCH INTERN

- 3D Scene Representation from Defocused Images
- 3D Human Avater Generation
- Mentor: Ho-Deok Jang

Seongnam, South Korea

Aug. 2024 - Feb. 2025

Military Service

REPUBLIC OF KOREA AIR FORCE

- Bird Alert Team (BAT) & Foreign Object Debris (FOD) Crew

Seongnam, South Korea

Sep. 2017 - Aug. 2019

PUBLICATIONS

First-Author Papers

[F6] SwiftVGGT: A Scalable Visual Geometry Grounded Transformer for Large-Scale Scenes

2026

JUNGHOO LEE, MINHYEOK LEE, SUNGHUN YANG, MINSEOK KANG, SANGYOUN LEE

Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR) Findings

[F5] CoMoGaussian: Continuous Motion-Aware Gaussian Splatting from Motion-Blurred Images

2025

JUNGHOO LEE, DONGHYEONG KIM, DOGYOON LEE, SUHWAN CHO, MINHYEOK LEE, WONJOON LEE, TAEHO KIM, DONGYOON WEE, SANGYOUN LEE

Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)

[F4] CoCoGaussian: Leveraging Circle of Confusion for Gaussian Splatting from Defocused Images

2025

JUNGHOO LEE, SUHWAN CHO, TAEHO KIM, HO-DEOK JANG, MINHYEOK LEE, GEONHO CHA, DONGYOON WEE, DOGYOON LEE, SANGYOUN LEE

Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)

[F3] SMURF: Continuous Dynamics for Motion-Deblurring Radiance Fields	2025
JUNGHO LEE, DOGYOON LEE, MINHYEOK LEE, DONGHYEONG KIM, SANGYOUN LEE <i>2nd Workshop on Neural Fields Beyond Conventional Cameras (CVPRW)</i>	
[F2] Hierarchically Decomposed Graph Convolutional Networks for Skeleton-Based Action Recognition	2023
JUNGHO LEE, MINHYEOK LEE, DOGYOON LEE, SANGYOUN LEE <i>Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)</i>	
[F1] Leveraging Spatio-Temporal Dependency for Skeleton-Based Action Recognition	2023
JUNGHO LEE, MINHYEOK LEE, SUHWAN CHO, SUNGMIN WOO, SUNGJUN JANG, SANGYOUN LEE <i>Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)</i>	
Conference Proceedings	
[C19] RTK-Guided Gaussian Splatting Pipeline for Georeferenced Urban 3D Reconstruction	2026
CHEOLHWAN KIM, WONJUN CHOI, YOUNGMOK KWON, JUNGHO LEE, MINHYEOK LEE, HONG-GYOO SOHN <i>International Society for Photogrammetry and Remote Sensing (ISPRS) Abstract-Only</i>	
[C18] Revisiting Weakly-Supervised Video Scene Graph Generation via Pair Affinity Learning	2026
MINSEOK KANG, MINHYEOK LEE, MINJUNG KIM, JUNGHO LEE, DONGHYEONG KIM, SUNGMIN WOO, INSEOK JEON, SANGYOUN LEE <i>European Conference on Computer Vision (ECCV)</i>	
[C17] MoRGs: Efficient Per-Gaussian Motion Reasoning for Streamable Dynamic 3D Scenes	2026
WONJOON LEE, SUNGMIN WOO, DONGHYEONG KIM, JUNGHO LEE, SANGHEON PARK, SANGYOUN LEE <i>Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)</i>	
[C16] SwiftVGGT: A Scalable Visual Geometry Grounded Transformer for Large-Scale Scenes	2026
JUNGHO LEE, MINHYEOK LEE, SUNGHUN YANG, MINSEOK KANG, SANGYOUN LEE <i>Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR) Findings</i>	
[C15] MonoCLUE: Object-Aware Clustering Enhances Monocular 3D Object Detection	2026
SUNGHUN YANG, MINHYEOK LEE, JUNGHO LEE, SANGYOUN LEE <i>The Association for the Advancement of Artificial Intelligence (AAAI)</i>	
[C14] GenCLIP: Generalizing CLIP Prompts for Zero-Shot Anomaly Detection	2025
DONGHYEONG KIM, CHAEWON PARK, SUHWAN CHO, HYEONJEONG LIM, MINSEOK KANG, JUNGHO LEE, SANGYOUN LEE <i>The 3rd Workshop in Vision-based Industrial Inspection (ICCVW)</i>	
[C13] Find First, Track Next: Decoupling Identification and Propagation in Referring Video Object Segmentation	2025
SUHWAN CHO, SEUNGHOO LEE, MINHYEOK LEE, JUNGHO LEE, SANGYOUN LEE <i>The 7th Large-Scale Object Segmentation (LSVOS) Workshop (ICCVW)</i>	
[C12] DepthFlow: Exploiting Depth-Flow Structural Correlations for Unsupervised Video Object Segmentation	2025
SUHWAN CHO, MINHYEOK LEE, JUNGHO LEE, DONGHYEONG KIM, SANGYOUN LEE <i>The 7th Large-Scale Object Segmentation (LSVOS) Workshop (ICCVW)</i>	
[C11] TransFlow: Motion Knowledge Transfer from Video Diffusion Models to Video Salient Object Detection	2025
SUHWAN CHO, MINHYEOK LEE, JUNGHO LEE, SUNGHUN YANG, SANGYOUN LEE <i>The 7th Large-Scale Object Segmentation (LSVOS) Workshop (ICCVW)</i>	
[C10] CoMoGaussian: Continuous Motion-Aware Gaussian Splatting from Motion-Blurred Images	2025
JUNGHO LEE, DONGHYEONG KIM, DOGYOON LEE, SUHWAN CHO, MINHYEOK LEE, WONJOON LEE, TAEHO KIM, DONGYOUN WEE, SANGYOUN LEE <i>Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)</i>	
[C9] CMTM: Cross-Modal Token Modulation for Unsupervised Video Object Segmentation	2025
INSEOK JEON, SUHWAN CHO, MINHYEOK LEE, SEUNGHOO LEE, MINSEOK KANG, JUNGHO LEE, CHAEWON PARK, DONGHYEONG KIM, SANGYOUN LEE <i>IEEE International Conference on Image Processing (ICIP)</i>	

- [C8] CoCoGaussian: Leveraging Circle of Confusion for Gaussian Splatting from Defocused Images** 2025
JUNGHO LEE, SUHWAN CHO, TAEHO KIM, HO-DEOK JANG, MINHYEOK LEE, GEONHO CHA, DONGYOON WEE, DOGYOON LEE, SANGYOUN LEE
Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)
- [C7] Effective SAM Combination for Open-Vocabulary Semantic Segmentation** 2025
 MINHYEOK LEE, SUHWAN CHO, **JUNGHO LEE**, SUNGHUN YANG, HEESEUNG CHOI, IG-JAE KIM, SANGYOUN LEE
Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR) - Oral Presentation
- [C6] SMURF: Continuous Dynamics for Motion-Deblurring Radiance Fields** 2025
JUNGHO LEE, DOGYOON LEE, MINHYEOK LEE, DONGHYEONG KIM, SANGYOUN LEE
2nd Workshop on Neural Fields Beyond Conventional Cameras (CVPRW)
- [C5] Video Diffusion Models are Strong Video Inpainter** 2025
 MINHYEOK LEE, SUHWAN CHO, CHAJIN SHIN, **JUNGHO LEE**, SUNGHUN YANG, SANGYOUN LEE
The Association for the Advancement of Artificial Intelligence (AAAI)
- [C4] Guided Slot Attention for Unsupervised Video Object Segmentation** 2024
 MINHYEOK LEE, DOGYOON LEE, SUHWAN CHO, CHAEWON PARK, **JUNGHO LEE**, SANGYOUN LEE
Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)
- [C3] Hierarchically Decomposed Graph Convolutional Networks for Skeleton-Based Action Recognition** 2023
JUNGHO LEE, MINHYEOK LEE, DOGYOON LEE, SANGYOUN LEE
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)
- [C2] Leveraging Spatio-Temporal Dependency for Skeleton-Based Action Recognition** 2023
JUNGHO LEE, MINHYEOK LEE, SUHWAN CHO, SUNGMIN WOO, SUNGJUN JANG, SANGYOUN LEE
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)
- [C1] Detection-Identification Balancing Margin Loss for One-Stage Multi-Object Tracking** 2022
 HEANSUNG LEE, SUHWAN CHO, SUNGJUN JANG, **JUNGHO LEE**, SANGYOUN LEE
IEEE International Conference on Image Processing (ICIP)

Journals

- [J5] Synchronizing Vision and Language: Bidirectional Token-Masking AutoEncoder for Referring Image Segmentation** 2026
 MINHYEOK LEE, DOGYOON LEE, **JUNGHO LEE**, SUHWAN CHO, SANGYOUN LEE
Pattern Recognition (PR)
- [J4] GenCLIP: Generalizing CLIP Prompts for Zero-Shot Anomaly Detection** 2025
 DONGHYEONG KIM, CHAEWON PARK, SUHWAN CHO, HYEONJEONG LIM, MINSEOK KANG, **JUNGHO LEE**, SANGYOUN LEE
Pattern Recognition (PR)
- [J3] Sparse-DeRF: Deblurred Neural Radiance Fields from Sparse View** 2025
 DOGYOON LEE, DONGHYEONG KIM, **JUNGHO LEE**, MINHYEOK LEE, SEUNGHOON LEE, SANGYOUN LEE
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- [J2] Treating Motion as Option with Output Selection for Unsupervised Video Object Segmentation** 2025
 SUHWAN CHO, MINHYEOK LEE, **JUNGHO LEE**, MYEONGAH CHO, SANGYOUN LEE
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- [J1] Multi-Scale Structural Graph Convolutional Network for Skeleton-Based Action Recognition** 2024
 SUNGJUN JANG, HEANSUNG LEE, WOJIN KIM, **JUNGHO LEE**, SUNGMIN WOO, SANGYOUN LEE
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)

Preprint

- OTT-Vid: Optimal Transport Temporal Token Compression for Video Large Language Models** 2026
MINSEOK KANG, MINHYEOK LEE, **JUNGHO LEE**, MINJUNG KIM, DONGHYEONG KIM, DAYEON LEE, HEESEUNG CHOI, IG-JAE KIM, SANGYOUN LEE
Submitted to *Neural Information Processing Systems (NeurIPS)*
- SCALE: Correcting Scale Drift with Contamination-Agnostic Loop Closure in Geometric Foundation-Model SLAM** 2026
CHUNGJAE CHOE, **JUNGHO LEE**, SEOKJUN LEE, NAKMYOUNG SUNG, CHANGICK KIM
Submitted to *The Association for the Advancement of Artificial Intelligence (AAAI)*
- Georeferenced Gaussian Splatting with RTK-GPS-Constrained Camera Pose Refinement for 3D Reconstruction** 2026
CHEOLHWAN KIM, WONJUN CHOI, YOUNGMOK KWON, **JUNGHO LEE**, MINHYEOK LEE, HONG-GYOO SOHN
Submitted to *Automation in Construction (AutoCon)*
- ESC-Net++: Effective SAM Combination for Open-Vocabulary Image and Video Segmentation** 2026
MINHYEOK LEE, **JUNGHO LEE**, SUHWAN CHO, SUNGHUN YANG, SANGYOUN LEE
Submitted to *Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*
- STATIC: Surface-normal-guided Two-branch Alignment and Temporal Integration for Video Depth Estimation** 2026
SUNGHUN YANG, MINHYEOK LEE, SUHWAN CHO, **JUNGHO LEE**, SANGYOUN LEE
Submitted to *Pattern Recognition (PR)*

PATENTS

Domestic Patent

- [P3] Palmpoint Recognition Method. KR-Application No.10-2023-0156996, Nov., 2023.
- [P2] Skeleton Graph-based Action Recognition Device and Method. KR-Application No.10-2023-0123693, Sep., 2023.
- [P1] Video Anomaly Detection Apparatus and Method using Relational Embedding. KR-Application No.10-2022-0156968, Nov., 2022.

PROJECTS

- Research on Robust Neural Rendering-Based Large-Scale 3D Ultra-Precision Virtual Space Generation and Spatial Registration for Low-Quality Noisy Data** *National Research Foundation of Korea*
DEEP LEARNING RESEARCHER *May. 2024 - Apr. 2027*
 - Development of large scene reconstruction by 3D Gaussian Splatting.
- Collaborative Perception and Intelligence Framework for Hyper-connected Interaction among Human and Intelligent Things** *Korea Electronics Technology Institute*
DEEP LEARNING RESEARCHER *Apr. 2024 - Dec. 2025*
 - Development of efficient skeleton-based action recognition model.
- Development of Anti-spoofing Model for Face Recognition Based on RGB Camera** *Samsung Electronics*
DEEP LEARNING RESEARCHER *Aug. 2023 - Jul. 2024*
 - Development of face anti-spoofing model robust to various spoofing attack.
- Development of Mobile Palmpoint Recognition Algorithm** *Samsung Electronics*
DEEP LEARNING RESEARCHER *Aug. 2022 - Jul. 2023*
 - Development of one-stage real-time mobile network, which includes keypoint detection and palmpoint recognition.
 - Development of real-time Android demo application for palmpoint recognition.
- Deep Learning-Based Initial Identification and Tracking System for Missing Persons in Heterogeneous CCTV Images** *National Research Foundation of Korea*
DEEP LEARNING RESEARCHER *Oct. 2018 - Dec. 2022*
 - Development of real-time multi-object tracking algorithm robust to occluded person.
- Development of AI Multi-Object Tracking and Behavior Analysis Technology** *Hanwha Techwin*
DEEP LEARNING RESEARCHER *Oct. 2020 - Oct. 2021*
 - Development of robust feature extractor for the object detection network.

Professional Services

Invited Talks

- “3D Neural View Synthesis from Degraded Images,” NAVER AI Lab Jul. 2025
- “3D Neural View Synthesis from Degraded Images,” Korean Electronics Technology Institute Jul. 2025
- “Leveraging Circle of Confusion for 3D Neural View Synthesis,” Korean Photonics Technology Institute May. 2025

Journal Reviewer

- IEEE Transactions on Graphics (TOG) 2025
- International Journal of Computer Vision (IJCV) 2024
- IEEE Transactions on Multimedia (TMM) 2024
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2023
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS) 2023
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) 2023

Conference Reviewer

- Association for the Advancement of Artificial Intelligence (AAAI) 2026
- IEEE/CVF Computer Vision and Pattern Recognition (CVPR) - Outstanding Reviewer (2025) 2025, 2026
- Conference on Neural Information Processing Systems (NeurIPS) 2024, 2025, 2026
- European Conference on Computer Vision (ECCV) 2024, 2026
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024
- The British Machine Vision Conference (BMVC) 2026

TEACHING EXPERIENCES

Understanding and Using AI

TEACHING ASSISTANT

Yonsei University
Spring 2022 - Fall 2025

Deep Learning Lab.

TEACHING ASSISTANT

Yonsei University
Spring 2023

Digital Logic Circuit

TEACHING ASSISTANT

Yonsei University
Fall 2021

SKILLS

Research and Development Stacks

- Main Languages** Python, C/C++, MATLAB, Kotlin
- Machine Learning** PyTorch, TensorFlow, Keras
- Computer Vision** OpenCV